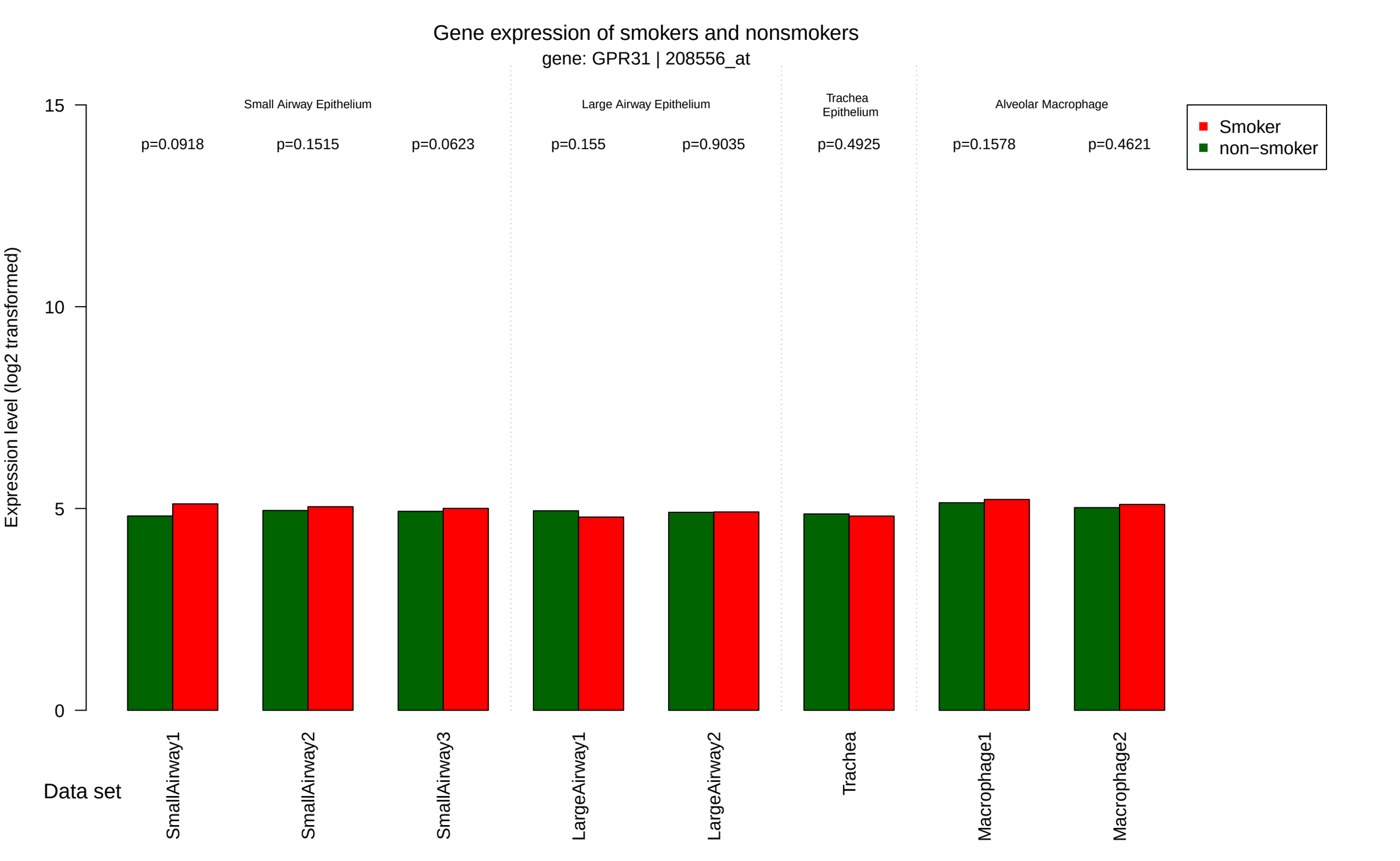
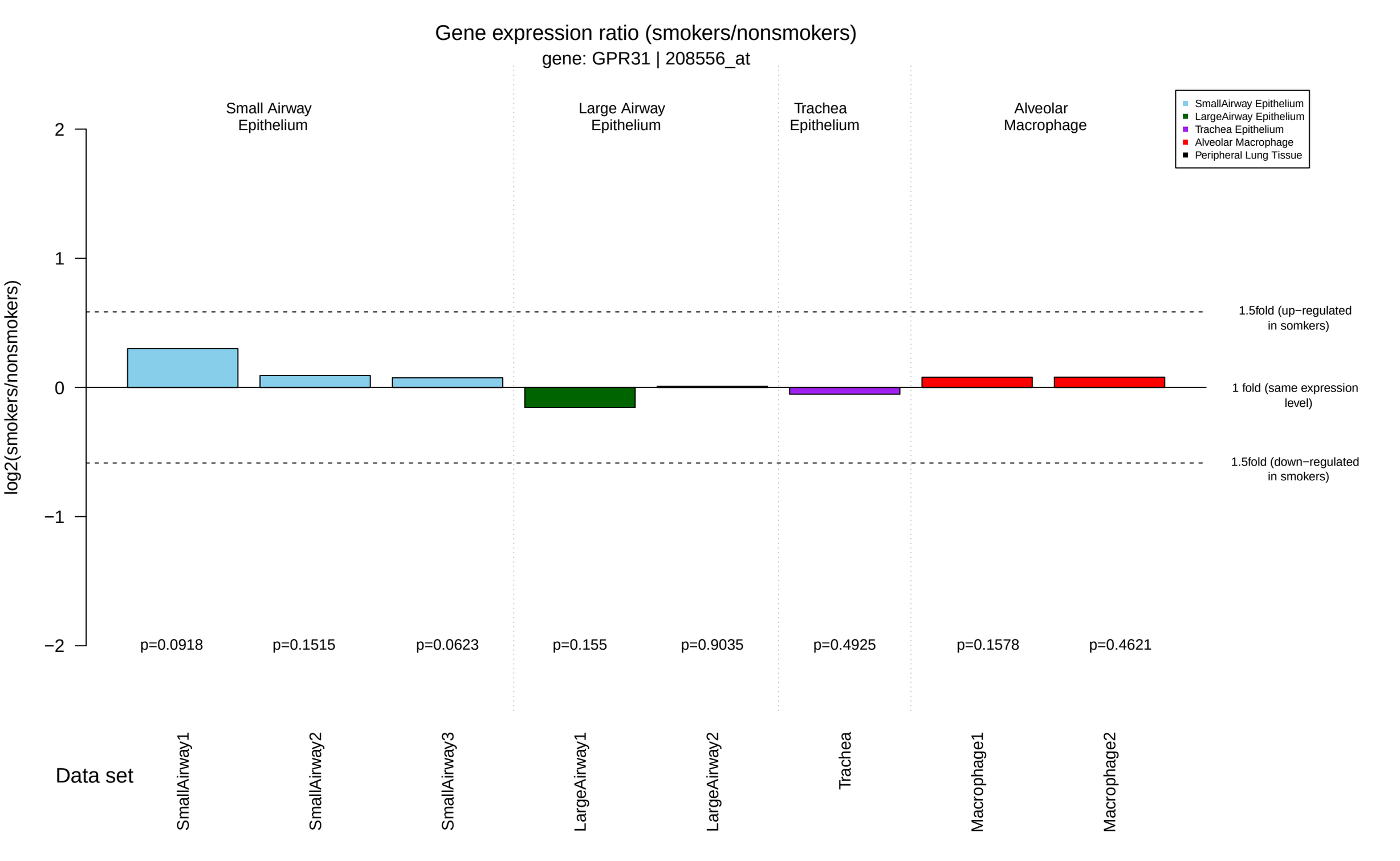
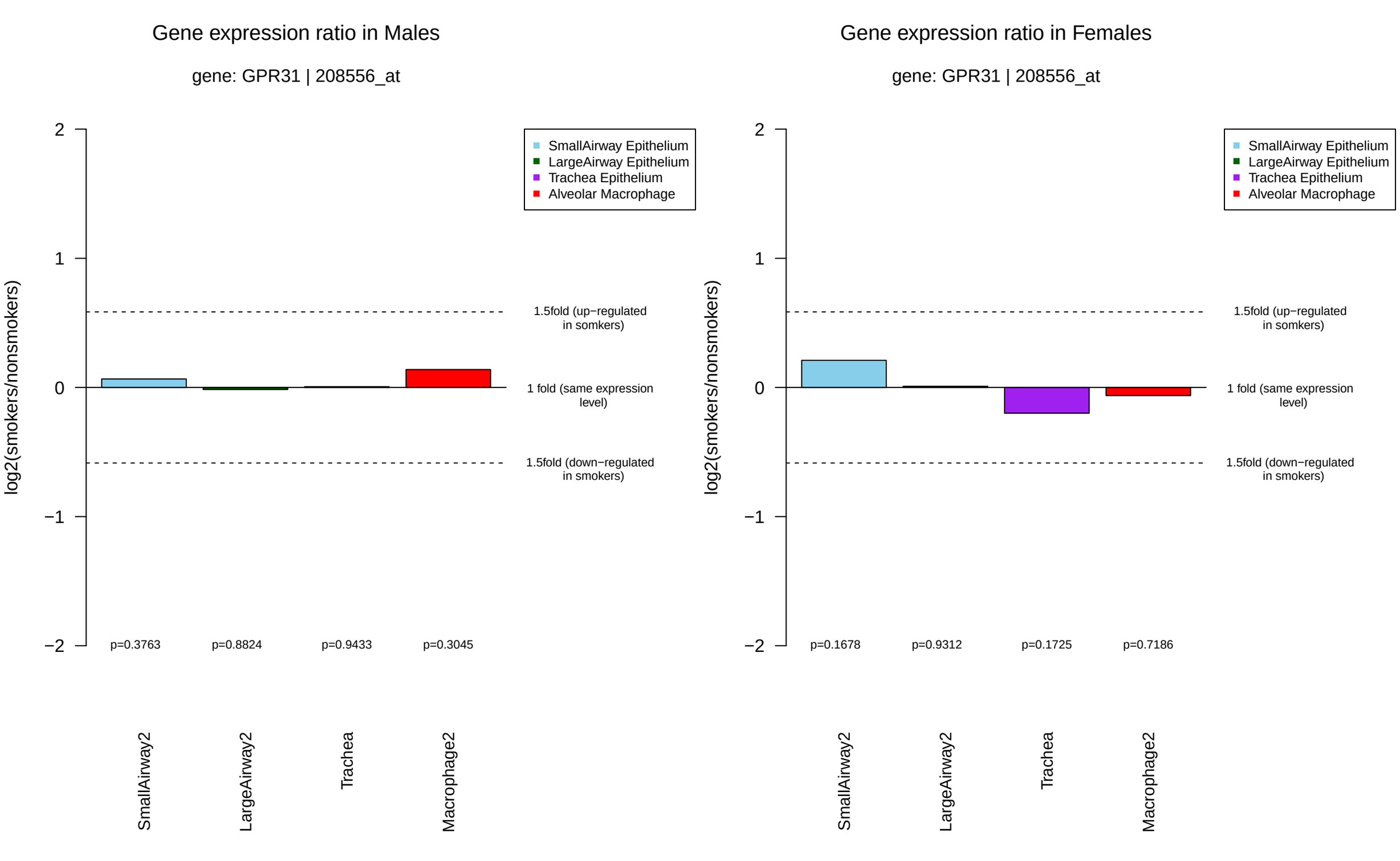
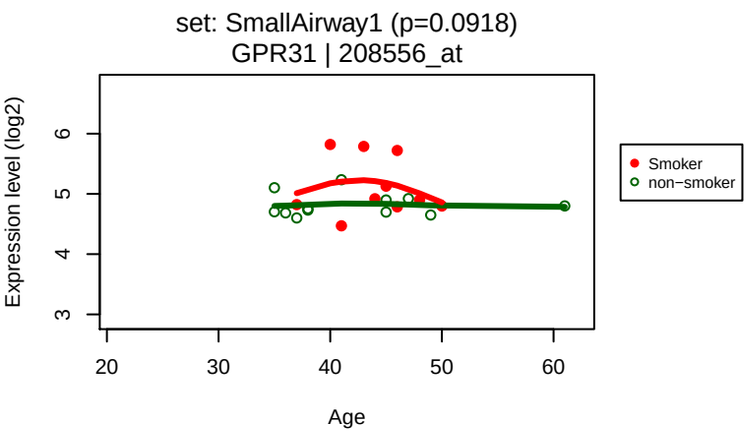
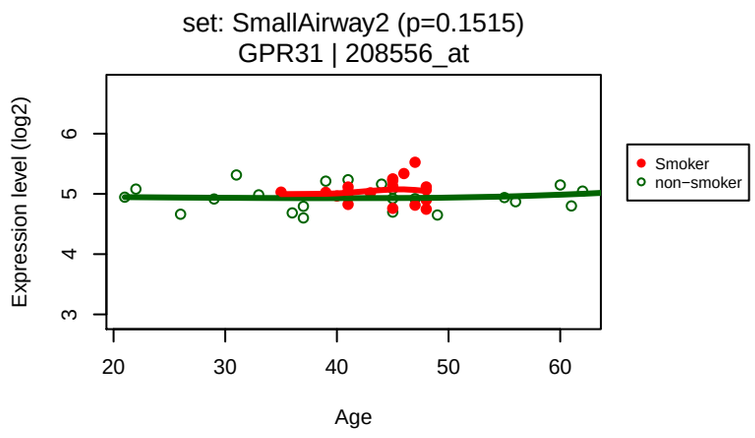
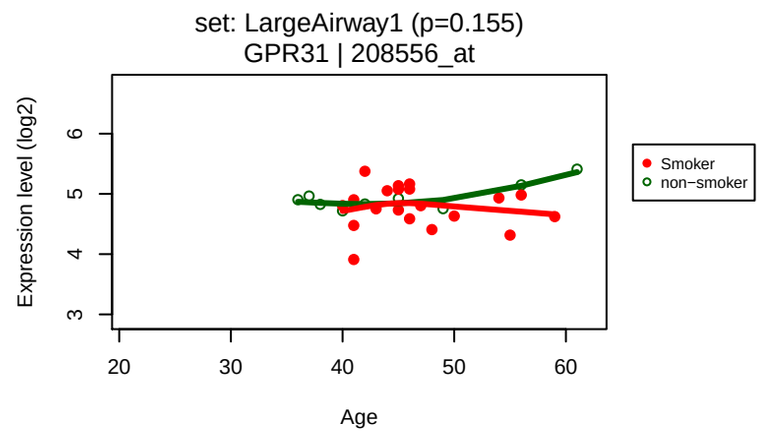

Gene expression of smokers and nonsmokers
gene: GPR31 | 208556_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0918
p=0.1515
p=0.0623
p=0.155
p=0.9035
p=0.4925
p=0.1578
p=0.4621
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

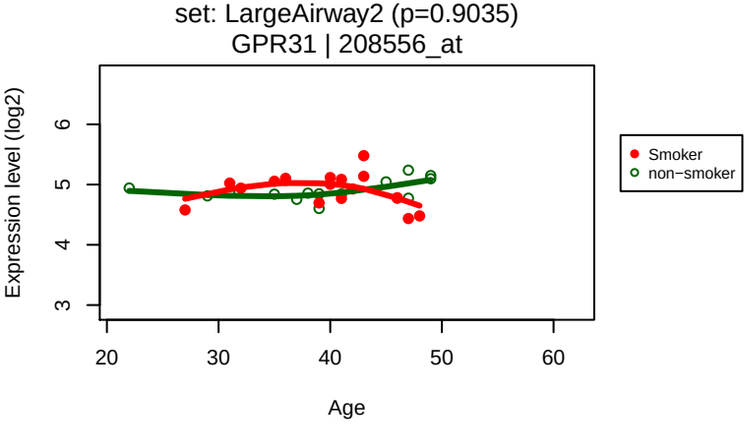

Gene expression ratio (smokers/nonsmokers)
gene: GPR31 | 208556_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0918
p=0.1515
p=0.0623
p=0.155
p=0.9035
p=0.4925
p=0.1578
p=0.4621
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

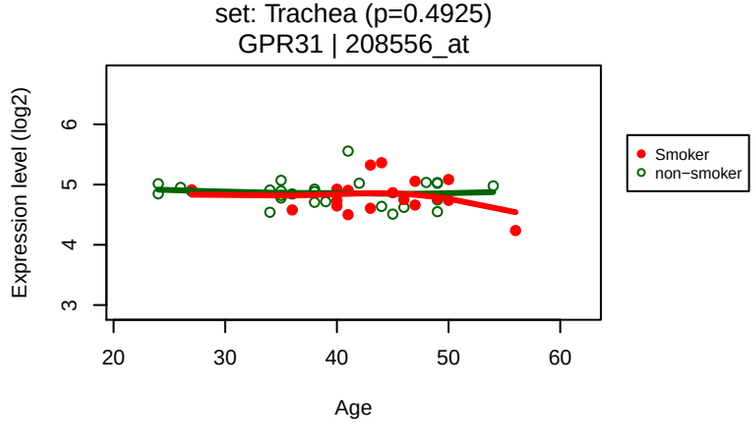

Gene expression ratio in Males
gene: GPR31 | 208556_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3763
p=0.8824
p=0.9433
p=0.3045
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GPR31 | 208556_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1678
p=0.9312
p=0.1725
p=0.7186
SmallAirway2
LargeAirway2
Trachea
Macrophage2

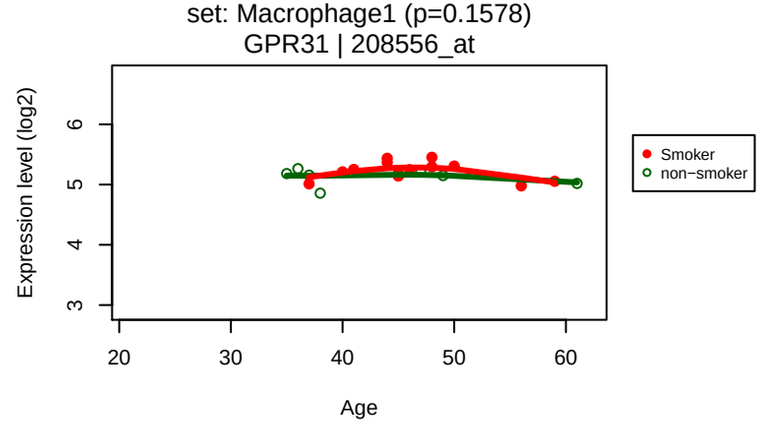

set: SmallAirway1 (p=0.0918)
GPR31 | 208556_at
Expression level (log2)
Age
Smoker
non−smoker

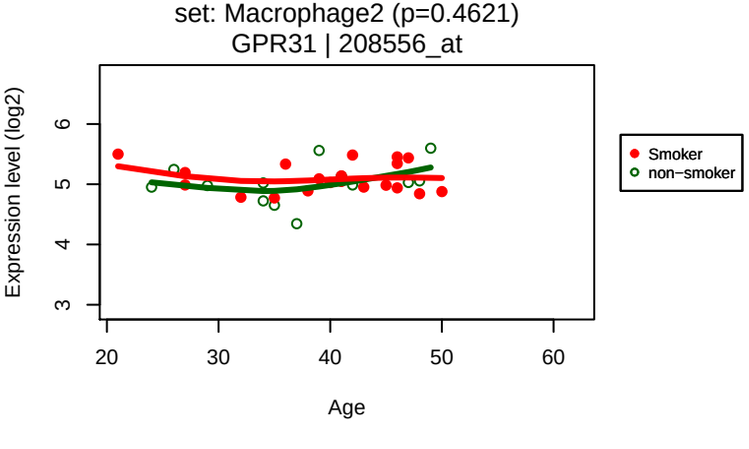

set: SmallAirway2 (p=0.1515)
GPR31 | 208556_at
Expression level (log2)
Age
Smoker
non−smoker

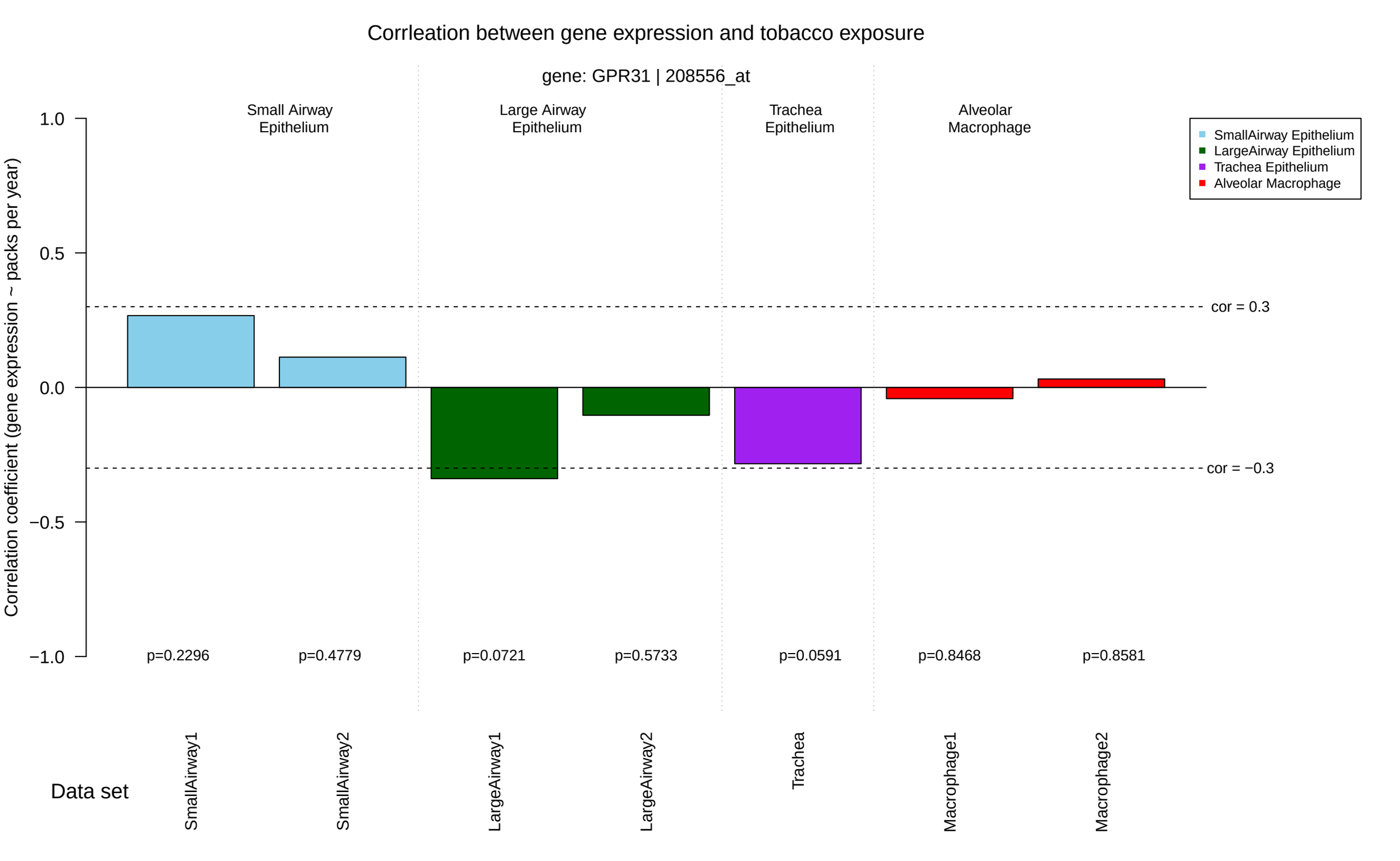

set: LargeAirway1 (p=0.155)
GPR31 | 208556_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9035)
GPR31 | 208556_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4925)
GPR31 | 208556_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1578)
GPR31 | 208556_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4621)
GPR31 | 208556_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GPR31 | 208556_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2296
p=0.4779
p=0.0721
p=0.5733
p=0.0591
p=0.8468
p=0.8581
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2